

All Plücker bodies determine the norm up to linear isometry

MathlibAnnex.EquivalentSeminorm.nonempty_linearIsometryEquiv_of_pluckerBodies_eq

theorem

Recovers a real linear equivalence preserving the two supplied norms.

Statement

Let $n = m + 1 > 0$ and $V = \mathbb{R}^n$ with the reference sup norm and standard Lebesgue measure λ . The continuous norms p, q have fixed positive comparison constants

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty.$$

Set $K_p = \{x : p(x) \leq 1\}$, $K_q = \{x : q(x) \leq 1\}$ and $v_p = \lambda(K_p)$, $v_q = \lambda(K_q)$, as positive finite real numbers. For a real linear map $A : V \rightarrow \mathbb{R}^N$, let $A_{ij} = (Ae_j)_i$ be its matrix in the standard bases. The minor-coordinate space has one coordinate for each increasing n -tuple I of distinct rows from $\{1, \dots, N\}$, and

$$\Pi_n(A)_I = \det(A_{I,*}).$$

Define the signed generator set and its real convex hull by

$$\mathcal{G}_p^N = \left\{ \sigma v_p \Pi_n(A) : \begin{array}{l} A : V \rightarrow \mathbb{R}^N \text{ is real linear, } \sigma \in \{1, -1\}, \\ \|Ax\|_\infty \leq p(x) \quad (x \in V) \end{array} \right\},$$

$$\mathcal{P}_p^N = \text{conv}_{\mathbb{R}} \mathcal{G}_p^N.$$

Define $\mathcal{G}_q^N, \mathcal{P}_q^N$ in the same way using q, v_q . The generators are vectors of minors, not linear maps. Assume $\mathcal{P}_p^N = \mathcal{P}_q^N$ for every N . There exists a real linear bijection $L : V \rightarrow V$ satisfying

$$\forall x \in V, \quad q(Lx) = p(x).$$

Assumptions

The two norm models and equality of every finite Plücker body are inputs.

Conclusion

The isometry is between V equipped with p and V equipped with q . Preservation of the reference sup norm is not the conclusion.

Proof route

Keep the same limit map and its ball image, recover the norms from positive scaled sublevel sets, and promote that map to a linear equivalence.

▼ Proof steps

1. **Choose the limit map once.** Input the two norm models and body equalities to limit recovery. It supplies bijective L , contraction, ball inclusion and the exact determinant-volume equation. ball image equality for that certificate gives $L(K_p) = K_q$ for the same L .
2. **Read both gauges from the ball image.** For each $t > 0$, homogeneity, linearity and bijectivity give

$$\begin{aligned} p(x) \leq t &\iff x/t \in K_p \\ &\iff L(x/t) \in K_q \\ &\iff q(Lx) \leq t. \end{aligned}$$

The middle equivalence uses $L(K_p) = K_q$. Its reverse implication also uses injectivity: if $L(x/t) = Ly$ with $y \in K_p$, then $x/t = y$. If the two nonnegative numbers $p(x)$ and $q(Lx)$ differed, their positive midpoint would satisfy one comparison and not the other. Thus $q(Lx) = p(x)$; at $x = 0$ both values are zero. The norm reconstruction from a bijective ball image takes positive definiteness of p, q , the real linear bijection and exact ball image as inputs, and returns this equality at each x .

3. **Retain that map in the equivalence.** Bijectivity promotes the same L to a real linear equivalence. Store it and $\forall x, q(Lx) = p(x)$ in the two fields of the model isometry structure. No sphere restriction is specified by these fields.

The recovered equivalence preserves p and q on the same coordinate vectors; it need not preserve the reference sup norm.

Main citations

- Exact declaration and source
- the signed generator set
- its real convex hull
- limit recovery
- ball image equality for that certificate
- norm reconstruction from a bijective ball image
- the model isometry structure

Lean source signature (exact)

```
/-- Equality of all finite Plücker bodies gives an equivalence preserving the stored norms. -/
theorem nonempty_linearIsometryEquiv_of_pluckerBodies_eq {m : ℕ}
  (MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ))
  (hBodies : ∀ N : ℕ, PluckerBody.body MX N = PluckerBody.body MY N) :
  Nonempty (LinearIsometryEquiv MX MY)
```

In the source	Mathematical meaning
<code>Fin (m + 1) → ℝ; →L[ℝ]</code>	The reference $V = \mathbb{R}^n$, $n = m + 1 > 0$, with sup norm; arrows mean continuous real linear maps. Source indices $0, \dots, n - 1$ correspond to formula indices $1, \dots, n$.

In the source	Mathematical meaning
<pre>MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ); MX.p; MY.p</pre>	The two continuous norms p, q and their positive comparison constants. The applications $MX.p\ x$ and $MY.p\ x$ mean $p(x), q(x)$.
<pre>MX.closedUnitBall; MY.closedUnitBall; MX.closedUnitBallVolume; MY.closedUnitBallVolume</pre>	The sets K_p, K_q and their real Lebesgue volumes v_p, v_q. Real volumes are the <code>toReal</code> of the finite extended measures.
<pre>hBodies : ∀ N : ℕ, PluckerBody.body MX N = PluckerBody.body MY N</pre>	For every nonnegative N, the real convex hulls $\mathcal{P}_p^N$ and $\mathcal{P}_q^N$ are equal. The source body is the convex hull of the vectors $\pm v_p \Pi_n(A)$, where $A : V \rightarrow \mathbb{R}^N$ is real linear and $\ Ax\ _\infty \leq p(x)$ for every x. The target body uses q, v_q in place of p, v_p.
<pre>Nonempty (LinearIsometryEquiv MX MY)</pre>	Existence of a real linear equivalence preserving the two stored norms, with the fields below.

From the two norm models and `hBodies`, the output `Nonempty (...)` asserts existence of one linear equivalence with the norm-preservation field below.

Related structure, shown separately: The norm-preserving linear-equivalence structure.

```
-- A linear equivalence preserving the two stored seminorms. -/
structure LinearIsometryEquiv (MX : EquivalentSeminorm E) (MY : EquivalentSeminorm F) where
  toLinearEquiv : E ≈l[ℝ] F
  map_p_eq : ∀ x, MY.p (toLinearEquiv x) = MX.p x
```

In the source	Mathematical meaning
<pre>toLinearEquiv : E ≈_l[ℝ] F</pre>	A bijective real linear map, together with its inverse. Here the two carriers are both V, and the map is the selected $L : V \rightarrow V$.
<pre>map_p_eq : ∀ x, MY.p (toLinearEquiv x) = MX.p x</pre>	$q(Lx) = p(x)$ for every $x \in V$. The source and target model parameters select which norm is applied on each side.

Surrounding assumptions and aliases.

▼ Exact content identity

Declaration: `MathlibAnnex.EquivalentSeminorm.nonempty_linearIsometryEquiv_of_pluckerBodies_eq`
Accepted content SHA-256: `f97501ad5aec2527ce2b2188acf6ce634ddb722f1fd641685d661e2a4050a3d0`
Accepted source guide SHA-256: `d8c3ceec1cea7b2596e3363e9ae717e5701a6330c73f051cfec5982b98c2f231`
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`